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Fuzzy Logic Based Prospects Identification System for Foreign Language Learning Through Serious Games

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ABSTRACT Interest in serious games for education has grown during the last years. The credit goes to the potential for engaging students with new ways to capture and maintain attention. The goal of this paper is to introduce a new multidisciplinary approach that incorporates psychological analysis theory with fuzzy inference and neural networks in the Foreign Language Learning prospects identification through serious games. Our research first used a Delphi method to accumulate Information Systems students' opinions resulting from a SWOT analysis of the use of serious games to learn English language. Then, we designed a Fuzzy Logic based Foreign Language Learning Prospects Identification System through serious games that takes four standard input parameters (Strengths, Weaknesses, Opportunities, Threats) already identified during our SWOT analysis and predicts the output (Learning Prospects of Foreign Language) by considering impact of certain variations in input parameters. Implementation results have been obtained through (MATLAB R2020a) and have shown reliable incites and findings.

INDEX TERMS Serious games, Foreign language learning, Delphi, investigative study, fuzzy logic, neural networks, inference system, adaptive neuro fuzzy inference system.

I. INTRODUCTION

In the 21st century, the technology use in the process of education has turned to be vital to empower the experience of learning. Students are now digital natives, learning knowledge and interpreting it differently [1]. Since birth, they have been familiar with the technology world, such as gaming, blogging, and social networking. As they spend most of their time on devices, they seem to reject traditional learning materials and feel bored very quickly in classrooms. The issue with digital natives is to attract their attention and engage them to learn [2]. To achieve this, professors from different disciplines, including teachers of English language, integrate gaming in classrooms as an innovative learning support to bring a touch of entertainment mixed with serious content [3]. Inserting serious games in classes will put students in harmony with what they are used to manipulating; mainly smartphones or computers [3]. In addition, students are no longer able to endure a stressful ambiance in classroom for a long period. Furthermore, they need to have an efficient and attractive educational system that meets their requirements.

Clearly, in computer-based ludic teaching and learning, serious games have become a genre of its own right [4]–[6]. As mentioned by the co-founder of the Serious Games Project, Ben Sawyer, ''serious games are serious business''. They are invading every single domain and could be extended to a wide range of application areas, such as government, military, corporate, healthcare and mainly education.

In the sphere of language learning, researchers are divided into two groups. On the one hand, some researchers consider game-based learning as not efficient and will not provide convincing results [7]. On the other hand, other researchers claim that computer-based games are quite convincing. For instance, a French research saw positive motivating results in a three-dimensional game of language learning [8].

Serious games are intended to smooth learning difficulties and bring an assistance in a soft manner specifically in crucial disciplines such as language learning [9]. According to

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researchers, the use of serious games strongly enhances students' performances [10]–[12]. However, researchers didn't analyze the strengths, the weaknesses, the opportunities and the threats of such new materials from students' point of view. They didn't analyze their needs from their own perspective.

To set up an essential awareness of serious games opportunities and weaknesses in learning foreign language, many questions should be addressed, including but not limited to: a) What could hinder or encourage serious games to be used as language learning teaching tools? b) What could motivate or prevent universities from spending money in serious games? c) What variables maximize the chance of serious games being good, and the high quality of results?

To our best knowledge, there was no suggestion for a taxonomy of outstanding challenges and opportunities resulting from the serious games use for learning a foreign language. As a result, in this research, a Delphi study was conducted to handle this gap. Delphi's scheme consists of a session of initial brainstorming, followed by an accumulation of main problems and a ranking of these matters. This culminated in a taxonomy of matters pertinent to English language learning game-based methods. In order to ensure the significance of the original brainstorming outcomes, we extended our conclusions by publicly addressing these issues. In specific, the current research has been investigative and experimental in nature and has established initial challenges and opportunities for serious games espousal.

This study will enable universities to improve their plan for serious educational games by understanding these new technologies' foremost challenges. The related problems found in this research also deliver an opening point for a deeper understanding and a further study of factors impacting the optimum utilization of serious games to learn English.

Further to explore the learning prospects of foreign language through serious games, we used Adaptive Neuro Fuzzy Inference System that was initially built in the early 1990s. Because it incorporates neural networks as well as fuzzy logic concepts, it can harness the advantages of both in one system. The inference method is compatible with a collection of ambiguous IF-THEN laws that are capable to learn and approximate nonlinear functions. Jyh-Shing Roger Jang from National Taiwan University, originally introduced ANFIS in 1993. It is a rigouous technique in soft computing that uses fuzzy logic to convert given inputs into ideal outputs by highly entangled elements of the Neural Network. The identified inputs (Strengths, Weaknesses, Opportunities, Threats) integrated in SWOT analysis may vary in certain situations, which may impact the learning capacity of an individual in constructive or destructive manner that is a stimulating task due to the inherent fuzziness of input data. These small variations in the values of input parameters can affect profoundly learning prospects based on the relation between input and output parameters. Therefore, to identify the relation between input and output parameters as well as to quantify the effects on output variable (Prospects of Learning Foreign Language through Serious Gaming) ANFIS is implemented.

The remainder of this study is arranged as follows. A literature review is presented in section 2. The methods to identify the opportunities and challenges connected with using serious games for learning English language are provided in section 3. Later, the outcomes of the analysis are detailed in section 4. Section 5 describes the proposed Fuzzy Logic based Foreign Language Learning Prospects Identification System through serious games that will take four standard input parameters (Strengths, Weaknesses, Opportunities, Threats) already identified during the SWOT analysis. With the assistance of a complementary qualitative analysis by an open dialogue with researchers, section 6 is devoted to exploring our observations and their consequences. The paper concludes with a description of significant findings, limitations, and pathways for further studies.

II. BACKGROUND

A. SERIOUS GAMES

Digital or video games are becoming an important part of our daily life. People around the world play games everywhere; when travelling, when relaxing and even at work. Indeed, 17% of the global population is joined in playing games [13]. Games are touching several domains and they are not restricted to the entertainment.

For instance, digital games with purposes are called "serious games". Serious Games designers exploit people's interest in digital games to draw their attention for a wide range of purposes that go beyond pure entertainment and claim that serious games have positive effects on the players' development of a number of different skills [14]. As a matter of fact, serious games allow players to experience situations that are difficult or even impossible in the real world for reasons of safety, cost, time, etc. [15].

The term "serious game" was initially utilized by Abt [16]. Several years later, this term was used by the US Army which released in 2002 the video game America's Army. During the same year, the Serious Game Initiative was founded by the Woodrow Wilson Center for Scholars in Washington, D.C. The main objective of the Serious Game Initiative is to use games as a dynamic technology to communicate cutting-edge research at the Wilson Center and beyond [14].

Michael and Chen define serious games as "games that do not have entertainment, enjoyment, or fun as their primary purpose" [17]. However, other authors consider the "entertainment" as a key component of serious games. One of these authors is Michael Zyda who defined serious games as "a mental contest, played with a computer in accordance with specific rules, that uses entertainment to further government or corporate training, education, health, public policy, and strategic communication objectives" [18].

Although different definitions were proposed in the literature, most of them agreed upon the fact that serious games are associated with the usage of gaming for drives other than mere entertainment. In the beginning, serious games have been suggested to instruct individuals to do tasks in specific jobs, such as instructing military officers. Then, they were progressed with the expansion of ''non-hard'' games and modern devices, including smartphones and tablets.

Since 2002, a wide range of serious games has been introduced for various purposes/aims, including: government, entertainment, healthcare, education, military and security, religion, corporate, culture and art, politics, environmental science, marketing, scientific research and many other flourishing domains. A recent report, entitled ''Serious games market by User Type, Application, and Industry Vertical: Global Opportunity Analysis and Industry Forecast, 2017-2023'', states that the global demand of serious games market was estimated to be \$2,731 million in 2016, and is expected to be \$9,167 million by 2023, rising from 2017 to 2023 at a CAGR 19.2%. The main factors driving the market growth are the increased need user engagement across enterprises, expansion of mobile-based educational games usage, and enhancement in learning outcomes. On the other hand, some threats delay the market growth, such as improper game design and lack of awareness about serious games [19].

Another perspective emphasizes the element of ''engagement'' in serious games. In fact, the authors of [14] consider that serious games are not just applying games and game technology for non-entertainment purposes, in domains such as education. They confirm that games should be motivating and engaging, which is a privilege for the development of different skills and abilities. Thus, serious games are clarified in [14] to be games which engage the user, and contribute to the implementation of a defined purpose other than pure entertainment whether or not the user is aware of it.

Serious gaming is considered to be a particularly active, problem-solving, situated and social mode of learning with fast and distinguished feedback that facilitates the enjoyment of learning [20]–[22]. Many academic studies affirm the optimistic impact of serious games on students' motivation [22] and performance [23].

Research concerning serious games' efficacy is not equivocal, and several researchers still epitomize their hype from gaming [24]–[26]. In this sense, researchers suggest that games are insignificant events that do not bestow to knowledge. Gosen and Washbush [27] express a skeptical position regarding the use of games in education. According to them, great work on methodology and on pedagogy should be prepared fitting a specific discipline. In addition, testing the game with students and requesting their feedback are mandatory.

Authors of [26] stress the importance of learning styles to be considered when developing educational games. According to them, a learning style is a determinant ingredient because learners have different requirements and personality traits, so their interaction with the same game will be completely different.

The mitigating views about learning experience and its efficiency conducted us to focus on the discipline of foreign language learning.

B. LANGUAGE LEARNING WITH SERIOUS GAMES

One of serious games' utmost possessions is its educational aspect. Indeed, they promote and open new horizons for active learning and provide an experience of learning by-doing. Serious games are well embraced by the young generation due to their graphics and design, as the serious games excel in keeping the young focused on their assignments and involving them in the learning process [28]. The game's learning aspect is the most significant. As ways to encourage student participate and practice, several discussions have explored reward systems. However, when people started to have a look at the games potential for student's engagement, they found that the games probably fit with the learning theories. Potentially, this approach will help students master the process of learning. The used integration between online gaming and education has a positive effect on student's learning thanks to their design and graphics, since serious games keep them concentrated on their tasks and engage them in the learning process [28].

It is crucial to learn languages, since it requires several competencies: speaking, listening, reading, writing, pronunciation, and dictation, which combines writing and listening. Language learning requires endurance and concentration, which are difficult to maintain for a long period of time. As a result, learners are expected to get bored easily, lose focus. Consequently, they will lack the learning engagement. Games offer a solution to this problem by the playful incorporation integrating difficult learning materials in a playful manner and make the process of learning encouraging and smooth [29]. The research related to serious games offers numerous language learning games, for example, English pronunciation [30] or German learning [31]. Some games are proposed for children with Down syndrome [32] to learn phonetics, etc. Existing games are considered to enhance students' performance and engagement. However, researchers did not analyze the overall process of learning with games. What should we stress and what should we avoid? What are the opportunities to exploit and what are the threats to overcome?

We need a close focus on these questions from students' point of view in order to deepen our understanding of the impact of serious games on foreign language learning.

Connection between gaming and learning should not be considered to be taken for granted because we will not see its irreversible impact only after a short period of time. The focus of this paper is on the effectiveness of serious games for learning foreign language. Indeed, researchers have identified prominent and exciting challenges and opportunities of serious games to learn foreign language. However, no nomenclature of such challenges and opportunities has been delivered. Thus, the literature in this study is providing the basics for such a taxonomy.

III. METHOD

To highlight the early-mentioned research gap, we clarified opportunities and difficulties our university might encounter

TABLE 1. Contributor's demographic data.

Study Data	
Total number of Contributors	49
Youngest contributors	24
Oldest contributor	27
Male contributor	42
Female contributor	7
Game experience	Often

when gaming is used in the scenario of learning foreign language. English language learning is essential for the curriculum of our students. Thus, our research has focused on English language learning as a foreign language, to illustrate our study. With students who belong to our university, we performed a SWOT analysis from a brainstorming session as a preliminary investigative research. The aim of this SWOT analysis is to deliver feasible perceived Strengths, Weaknesses, Opportunities, and Threats of using serious games in learning English language.

The SWOT analysis created a systemized set of features that could enable Universities to get deeper awareness of the benefits and costs of serious games in English language learning. These features could benefit to notify the staff members about the pros of working in a game-based setting, as well as increase their understanding of the potential threats innate to serious games use. The following questions are considered to address SWOT analysis categories:

- *Strengths:* Which factors are considered to be main advantages of serious games?
- *Weaknesses:* Which factors are considered to be main disadvantages of serious games?
- *Opportunities:* What potential chance is offered by serious games with regards to learning English language?
- *Threats:* What are the potential threats pose by serious games with regards to learning English language?

A Delphi approach is used to conduct our exploratory study. Delphi studies are regularly used in information system studies when there is a need to achieve a consensus among field specialists on a topic where idea formation is required [33].

A. THE CONTRIBUTORS

Students from our university, precisely enrolled in the College of Computer and Information Sciences, were asked to contribute to this study. They were required to do a SWOT analysis of serious games and gamification in learning the English language. The students' experience with serious games differ from a student to another. As a matter of fact, some students participating in our study have participated to use serious games in certain disciplines such as business. However, other students are not familired with the concept of serious games.

During the brainstorming process, we asked the contributors to use Duolingo as English learning serious game for a period of time before proceeding to the SWOT analysis.

As clarified in Table 1, demographics of the research contributors have been provided.

B. THE BRAINSTORMING PROCESS

The thinking procedure comprised of various tasks in which the contributors were required to participate during a span of 360 minutes. This is accomplished by a research process and a summarized agenda as follows:

- After a preliminary presentation on the use of "serious" gaming, students were required to namelessly create ideas about the four themes of SWOT.
- Later, students were divided into four subgroups and they were required to clarify, reduce and establish jointly created concepts into distinguished statements about one of the SWOT four themes. Each subgroup was evaluated by a facilitator in this task. The purpose was to unite on alike concepts, eliminate nonrelated ones, and rewrite those inadequately understandable.
- Students then joined each other as a single group. Facilitators of the subgroups introduced and explained the selected statements to the group for their particular stuff.
- Then students were required to rate the appropriateness of each remark on a 10-point Likert-type scale independently and anonymously, with '10' being a very related statement and '1' being the least related statement referring to each of the four SWOT variables.
- After that, the scores of voting were introduced in a raw arrangement to all contributors for a conversation stimulation of the outcomes (suggestion by suggestion) and, where possible, to facilitate the regeneration of suggestions, to explain the standard deviations of the ratings and thereby to create a cooperative agreement.

We held an open debate with the same students about the thinking outcomes for 90 minutes to complete this process. We tried to deepen and explain their interpretation of the meaning of the statements' influence and their contribution in making serious games-based English language learning effective. To facilitate the discussion, a list of controlling interrogations managing these concepts was ready.

C. SWOT ANALYSIS

Students generated a total of 167 ideas during the initial session of brainstorming, which aimed at defining the four components of SWOT analysis. Later the proposals were reshaped and revised, then reclassified as defined in the process of brainstorming. The statements have been lowered to 41 (10 weaknesses, 12 strengths, 5 threats and 14 opportunities). In order to increase the accuracy of the statements, after data collection, the total findings of the SWOT analysis were checked and paraphrased. Finally, in order to ensure that the ideas are properly represented and categorized, we asked the contributors to answer some important questions regarding the statements.

- Are the resultant statements possibly miscategorized between opportunities and strengths or between threats and weaknesses?
- Are the resultant statements suitable and within the serious games scope?

TABLE 2. Strengths consolidated from an initial set of 53 ideas and then reviewed, reclassified, and paraphrased.

Strengths	Mean	SD
Welcome break from the common routine of the language class	7.95	0.55
More motivation, challenge and excitement	9.2	0.71
Making and supporting the struggle of learning	8.65	0.94
Providing language practice with various skills; writing, speaking, reading and listening	8.73	0.66
Encourage students to interact and communicate and have fun	9.57	0.49
Refining the receptiveness of the students.	8.08	0.76
Motivating the student's instinct for attainment	8.69	1.33
Keeping all the involved and interested students	8.34	1.03
Creating a meaningful context for language (concentration on the language use rather than on the language itself)	9.0	0,24
Offering the chance to learn, practice, or review explicit language stuff	8.0	1.14
Better students' work	9.1	0.43
Environment personalization and flexibility	9.12	0.82

- Are the resultant statements special to a specific serious game?

After deep examination, the results are organized and put into four tables, (SD = Standard Deviation).

Contributors identified a list of interesting strengths as results of the brainstorming session presented in Table 2. The findings are aligned with researchers [22], [28]–[31]. First of all, contributors shed light on the significance of gaming procedure as a different way from the typical routine followed in the language class. They focused on the substantial feature of such classes ''heavy content to apprehend and to apply, games could bring fluidity to this type of classes''. They highlighted the link between ludic environments and motivation to learn. They cited that such learning materials ''will bring a challenging, motivating and exciting experience of learning'' which will sustain the effort of learning that meets the work of [28]. Ludic process of learning encourages students ''to interact, communicate and have fun in the same time'' which will increase students' responsiveness and concentration and stimulate their achievement instinct. Consequently, ludic learning process will keep students involved and interested which will rise their engagement to learn. It will allow them to ''practice language with various skills such as reading, listening, writing and speaking'' which meets the work of [29]–[31]. It will provide them with the opportunity to learn, practice or review specific language material which will increase their performance and urge them to focus on the use of the language. Furthermore, students affirmed that games provide the possibility to customize the game environment. One contributor said, ''personalization allows us to be identified with our own profile'', i.e., to be different from others.

According to the contributors, many weaknesses seem to pervade language learning by the means of serious games as shown in Table 3. Findings of this section are meeting the hypes of researchers [24]–[27]. First, several students will not feel comfortable because of the competitiveness levels

TABLE 3. Weaknesses consolidated from an initial set of 43 ideas and then reviewed, reclassified, and paraphrased.

Weaknesses	Mean	SD
Comfort Lack because of various competitiveness levels between students.	7.12	0.99
Frustration resulted from the game misunderstanding.	5.69	3.53
Inadequacy of the game with the general class vibe.	6.36	1.46
Control lack of the class because of the ludic ambiance	4.51	1.66
Discipline issues because of the ludic general ambiance	4.69	2.02
Distraction from the game real goal without learning	8.02	0.56
Precepting learning with games as unnecessary, time waste, immature and childish	4.16	2.08
Cheating and fraud	8.61	1.12
Loss of touch with reality	6.12	1.93
Requiring many technical resources	3.42	1.42

TABLE 4. Opportunities consolidated from an initial set of 50 ideas and then reviewed, reclassified, and paraphrased.

Opportunities	Mean	SD
Unremitting self-learning	9.24	0.92
Nonstop assessment of the students	9.08	0.6
Complete liberty in gaining knowledge	9.57	0.49
No gender or social unfairness	9.59	0.48
Margin expanding of freedom to error without any negative repercussions	9.85	0.25
Joy and fun increase in the classroom and creating great ambiance for learning	8.36	0.76
Learning by different educational means	9.18	0.73
Linking the education with the practical application and real life	8.53	0.93
Offering an unlimited and appropriate set of tasks for students	9.08	0.52
Inspiring students to find out their learning motivation	8.18	0.64
Engagement of students	5.61	2.06
Redirecting the interests of students	6.02	1.21
Making education meaningful and leave an outcome for the pupil	8.30	1.31
Automated Learning	9.44	0.65

difference between them. They might get frustrated if the game is misunderstood by them. It may also be ''not adequate for the general class vibe and provoke a kind of distraction''. Students argued that the ludic atmosphere ''may generate a lack of control on the class and lead to discipline issues''. They stressed ludic environment may lead them to be distracted from their real goal without learning. They ''may be attracted to the game without achieving a real goal''. Students highlighted that the use of games in classrooms may make them perceive learning as childish, immature, a waste of time or unnecessary. According to contributors, games may lead to losing touch with reality. Finally, games may require many technical resources and may lead to increase cheating and fraud if the internet access is granted on the device.

The contributors identified many relevant opportunities listed in Table 4 that may help us to better understand the use of games in classrooms. The majority of the statements have high means with a low SD. The findings of this section were mainly highlighted by [28]–[31]. First, they confirmed that games provide the possibility to have a continuous self-learning and assessment with unlimited and appropriate set of tasks. Consequently, students will have the complete liberty in gaining knowledge. Students appreciate how games expand the margin of liberty to error without any undesirable consequences. This ''ensures to everybody to express him/herself more easily and freely''. In addition, they argue that ''they will not receive any gender or social discrimination which will have a relevant consequence on students' performance and self-esteem''. Games will bring fun, joy and create a pleasant ambiance for learning. They will make learning possible with new evolving means which may attract students. According to the contributors, games may redirect the interests and the attention of students. They may inspire students to find out their learning motivation. One student said ''games may make us very attentive and more concentrate so better engaged to what we are doing''. Another student stresses that ''games may make the educational procedure more meaningful and leave better effect on our emotional state so courses will be an instant of enjoyment without pressure or tediousness''. Lastly, the contributors mentioned that game will provide the access to an ''evolved automated learning more coherent with new generations''. According to them, various recent educational supplies e.g., blackboard, should be replaced with smarter software or equipment (game).

In brainstorming activities, the last phase was linked to threats of games as learning stuff as shown in Table 5. In fact, threats on class values and marks degradation are highlighted by many contributors. They argued that ''games may bring negligence and distraction'' [24], [25], [27]. Students affirmed that in this kind of learning context, social loafing is hard to regulate. One contributor affirmed, ''The professor could not supervise the activity of all students in the same time, so we don't know if they are exercising or doing something else''. Second, students have put the spot on the impact of the misunderstanding of the game on their behavior. According to them, games ''could bring a motivation if the students lose frequently and did not apprehend the content if the game is presenting incoherent tasks (i.e., hard tasks for beginners)''. In addition, games may bring ''boredom and lack of involvement if the learner finds that the content is strange or inadequate'' (non-adequate learning style/content). Students confirmed that the provided game must consider learner language level, his/her major gaps, his/her mother tongue and his/her gender (customization of the game). Finally, students argued that the over usage of games in education could have a harmful effect on students as they may cause strong dependency on machines. For instance, courses may be postponed or canceled if the game presents any dysfunctions or bugs. They stressed that games will never replace a teacher.

Discussing the strengths, weaknesses, opportunities, and threats of serious games during the brainstorming activity yielded results that appeared to relate to factors arguably relevant for foreign learning using serious games, such as motivation to learn, customization, learning style, engagement. These are factors which distinctly influence foreign language learning with serious games. They are neither opportunities nor threats but fundamental constructs that impact learning. These constructs were mentioned in the literature as stated in Table 6.

TABLE 5. Threats consolidated from an initial set of 21 ideas and then reviewed, reclassified, and paraphrased consolidated.

Threats	Mean	SD
Degrading marks due to negligence	5.61	1.96
Degrading class values	5.24	1.49
Amotivation caused by excessive loosing.	7.36	1.29
Boredom and involvement lack in learning	5.69	1.77
High dependency on machine reliability	7.44	1.74

D. PROSPECTS IDENTIFICATION THROUGH ANFIS BASED SIMULATION FOR ENGLISH LANGUAGE LEARNING USING SERIOUS GAMES

Eventually an Adaptive Neuro Fuzzy Inference System (ANFIS) based expert system is designed to predict overall impact of serious games on learning English as foreign language. Expert systems possess human-like expertise for data analysis and decision making. We used it in a setting where, a high degree of vagueness exists.

The Fuzzy Logic (FL) technique is comparatively complex as it requires linguistic rules mapping for information processing as in human's cognitive process [42]. But (ANFIS) handles this complexity very well by using hybrid mechanism contains Fuzzy Logic and Artificial Neural Networks. Therefore, ANFIS is more advantageous for further purification of identified prospects through eradication of the fuzziness incurred by the input parameters (Strengths, Weaknesses, Opportunities, and Threats) which are integral components of SWOT analysis [43]. Though SWOT analysis is a very advantageous technique to analyze the dataset but it presents the information about parameters discretely without any association among them i.e., it can quantify the strengths, weaknesses, opportunities and threats but it will be difficult to quantify the overall impact of the mentioned four parameters on the output. The ANFIS is able to design an input-output mapping based on both human intuition (in the form of fuzzy If-Then rules) and specified input-output pairs with the use of hybrid learning mechanism. The ANFIS have the ability to simulate nonlinear functions, by learning patterns of nonlinear systems through dataset presented for training and testing. It already shown impressive results in different simulated scenarios.

The Takagi–Sugeno approach is currently the best-known paradigm for applying ANFIS. This definition of model uses the fuzzy generic IF-THEN rules, similar to other fuzzy methods. This nonlinear model simulates the system's input/output connections, limit the dynamics of each fuzzy rule by constructing a linear model of the system. The subsequent fuzzy rules will define a general Takagi–Sugeno based FIS model for nonlinear systems [44].

If $z_1(w)$ is F_{i1} And If $z_2(w)$ is F_{i2} And ...... If $z_p(w)$ is F_{ip} And Then refer to (1) and (2);

$$\begin{cases} x(w) = (D_{i+}\Delta D_i).x(w) + (E_i + \Delta E_i).y(w) \\ y(w) = (G_i + \Delta G_i).x(w) \end{cases} \tag{1}$$

where the notations have the following meaning:

- $z_p(w), j = \{1, 2, \ldots, p\}$

  are the propositional varialbles.

- $F_{ij}(w), i = \{1, 2, \ldots, R\}$ and $j = \{1, 2, \ldots, p\}$

  depicts the fuzzy varialbles.

- R is the number of defined fuzzy rules;
- $x(w) \epsilon V^n$ is the state vector.
- $y(w) \epsilon V^m$ is the input vector.
- $D_i \epsilon V^{n*n}$ is the state matrix.
- $E_i \epsilon V^{n*m}$ is the input matrix.
- $G_i \epsilon V^{q*n}$

  is the output matrix where q is the number of output parameters.

- $\Delta D_i, \Delta E_i, \Delta G_i$

  are the matrices that in the uncertainities.

The above-mentioned assumptions lead to the inferred model below:

$$\begin{cases} x(w) = \sum_{i=1}^{r} t_i(z(w))(D_{i+}\Delta D_i).x(t) + (E_i + \Delta E_i).u(w) \\ y(w) = \sum_{i=1}^{r} t_i(z(w))(G_i + \Delta G_i).x(w) \end{cases} \tag{2}$$

where $t_i(z(w))$ is the normalized grade of membership for each rule and is compliant with (3).

$$\begin{cases} \sum_{i=1}^{r} t_i(z(w)) = 1 \\ 0 < t(z(w)) < 1 \end{cases} \tag{3}$$

These equations signify the essentials of the execution of an optimized method for the special case of prospects identification.

In this research, we have used Sugeno-Type Fuzzy Logic based technique for further analysis of the response generated through SWOT analysis to predict the behavior of students for learning foreign language by means of serious games. For this purpose, Fig. 1, shows the variables included in SWOT analysis as input variables and overall Impact of serious games on Foreign Language Learning as Output variable using Sugeno Model.

ANFIS deployed to eliminate the inherent fuzziness of patterns gained through SWOT analysis for the exact identification of i.e., Neutral, Negative Impact, Positive Effect or Highly Influential. Fig. 1 illustrates the Fuzzy Inference System (FIS), and then membership functions will be defined for all input parameters related to the Serious Games' Impact on learning foreign language, as shown in Table 7. Then, a rule base is generated for the identified parameters displayed in Table 8. The initial configuration of ANFIS is presented in Table 9. We have used 70% of the dataset as training data, kept 15% for testing, and finally 15% for checking. Finally, the precisely identified prospects can be seen through the different surface views.

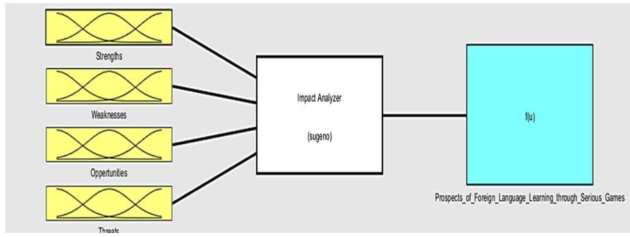

FIGURE 1. **ANFIS based impact analyzer's input and output variables.**

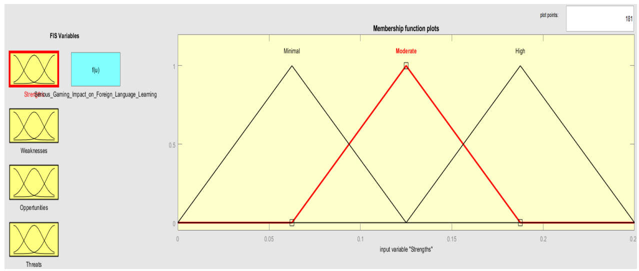

FIGURE 2. **"Strengths" variable's membership function.**

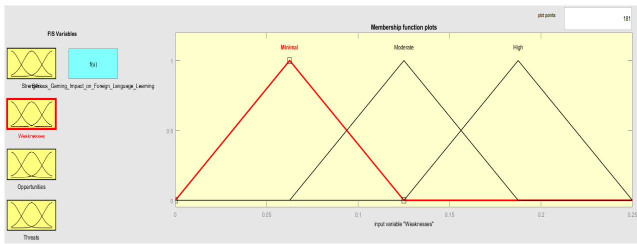

FIGURE 3. **"Weaknesses" variable's membership function.**

Fig. 2, shows the Strengths variable's membership functions for impact on foreign language learning i.e., Minimal, Moderate and High.

Fig. 3, shows the Weaknesses variable's membership functions for impact on foreign language learning i.e., Minimal, Moderate and High.

Fig. 4, shows the Opportunities variable's membership functions for impact on foreign language learning i.e., Minimal, Moderate and High.

Fig. 5, shows the Threats variable's membership functions for impact on foreign language learning i.e., Minimal, Moderate and High.

In Fig. 6, we have defined four membership functions Neutral, Negative Impact, Positive Effect and Highly Influential of defined output variable "Serious Gaming Impact on Foreign Language Learning" by considering all input variables and the fuzziness they incurred.

TABLE 6. Constructs impacting English learning using serious games.

Construct	Definition and Categories	Reference
Motivation to learn	Two main categories: Extrinsic and intrinsic motivation. • Intrinsic motivation refers to the fact to do an activity for itself, and the pleasure and satisfaction derived from participation. It has different types such as intrinsic motivation to know, intrinsic motivation toward accomplishments and intrinsic motivation to experience simulation. • Extrinsic drive relates to a wide variety of behaviors which are involved in as a sources to an end and not for their own sake. It could be seen as external regulation, integrated regulation or introjected regulation.	[34]
Customization (in virtual environments and games)	Customization is the opportunity to personalize and bring a personal touch to the game environment. It has three main categories: • Functional customization: Customization that directly impacts the dynamics and mechanics of the game and therefore has a direct impact on individual player gameplay. Customizing character skills is an example of this type of customization. • Cosmetic customization: Customization that does not impact on game dynamics and mechanics. Avatar customization is an example of this type of customization. Although it does not directly impact gameplay, it might impact a player's enjoyment of the game. • Usability: Customization that does not directly impact dynamics and mechanics of the game but might impact player performance, such as interface customization. It may have an effect on players' gameplay experience.	[35] [36]
Engagement (in education)	Engagement is comprised of behavioral (participation in class and school) and affective components (school identification, belonging, valuing learning). It has two sub dimensions: • Cognitive: value of learning, and personal goals and autonomy. • Psychological engagement: feelings of identification or belonging, and relationships with teachers and peers.	[37][38]
Learning style	The manner in which they characteristically approach learning tasks. Learning styles can be categorized along 4 dimensions namely: • Sensing/Intuitive: relates to how a student perceives the world according to their abstraction management skills. • Visual/Verbal: differentiates students who are visually orientated from students who are verbally orientated.	[39] [40] [41]

In Tab. 5 we have shown some mathematical representation of membership functions that have been used to assign linguistic variables for each incoming input during fuzzification, as well as output during defuzzification process.

We want to analyze the rationality of the defined input variables by identifying their impact on output variable.

TABLE 7. Mathematical representation of membership functions.

Input Parameter	Member Functions
Strengths ($\mu_{Strengths(s)}$)	$\mu_{strengths,minimal(x,0,013,2)=exp}\left[-1/_2\,\lvert x-0/_{0.13}\rvert^2\right]$
	$\mu_{strengths,moderate(x,0.13,024,2)=exp}\left[-1/_2\,\lvert x-0.13/_{0.24}\rvert^2\right]$
	$\mu_{strengths,high(x,0.24,025,2)=exp}\left[-1/_2\,\lvert x-0.24/_{0.25}\rvert^2\right]$
----------------	--
Weaknesses ($\mu_{Weaknesses(w)}$)	$\mu_{weaknesses,minimal(x,0,0.13,2)=exp}\left[-1/_2\,\lvert x-0/_{0.13}\rvert^2\right]$
	$\mu_{weaknesses,moderate(x,0.13,0.24,2)=exp}\left[-1/_2\,\lvert x-0.13/_{0.24}\rvert^2\right]$
	$\mu_{weaknesses,high(x,0.24,025,2)=exp}\left[-1/_2\,\lvert x-0.24/_{0.25}\rvert^2\right]$
Impact ($\mu_{Impact(i)}$)	$\mu_{impact,neutral(x,0,0.25,2)=exp}\left[-1/_2\,\lvert x-0/_{0.25}\rvert^2\right]$
	$\mu_{impact,negative\ impact(x,0.25,0.50,2)=exp}\left[-1/_2\,\lvert x-0.25/_{0.50}\rvert^2\right]$
	$\mu_{impact,negative\ impact(x,0.50,0.75,2)=exp}\left[-1/_2\,\lvert x-0.50/_{0.75}\rvert^2\right]$
	$\mu_{impact,negative\ impact(x,0.75,1,2)=exp}\left[-1/_2\,\lvert x-0.75/_{1}\rvert^2\right]$

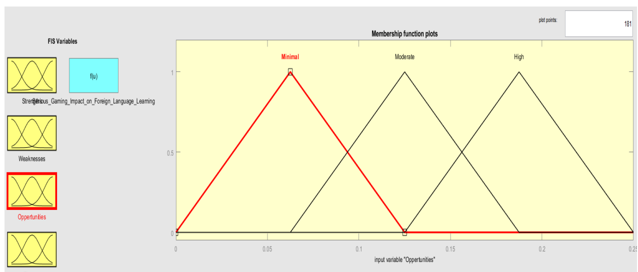

FIGURE 4. "Opportunities" variable's membership function.

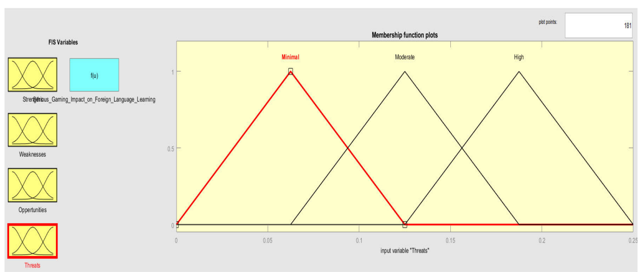

FIGURE 5. "Threats" variable's membership function.

To analyze this, we have initially generated 13 rules while the rest of the rules which are 81 have been defined during training by the fuzzy system after learning the patterns

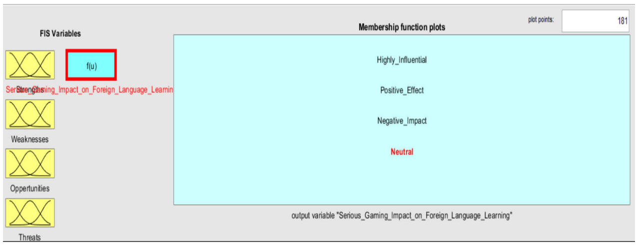

FIGURE 6. "Serious gaming impact on foreign language learning" variable's membership function.

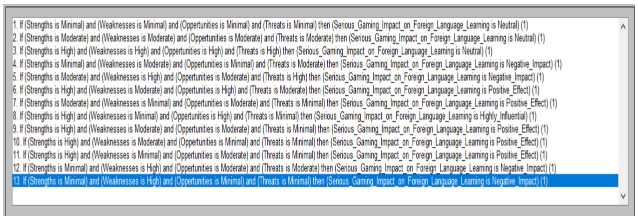

FIGURE 7. Preliminary defined rules for overall impact identification.

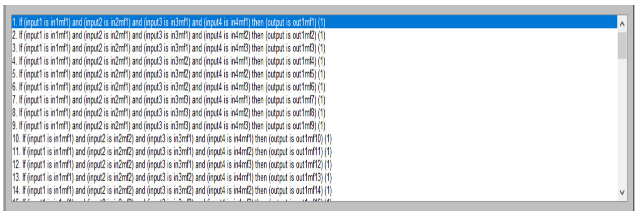

FIGURE 8. Rules defined during training process.

of dataset as it happen in human cognitive process, shown through Fig. 7 and Fig. 8 respectively.

Fig. 9 to Fig. 12 show the graphical representation of defined rules that clarify how the input variable variations influence the output variable.

Fig. 13 represents the post-training phase rule viewer based on the identified input variables. We can analyze the impact on output. We have 81 rules now for analyzing the impact of serious games on learning foreign language based on membership functions of strengths, weaknesses, opportunities and threats. Multiple impacts are shown with the changing of input values.

Fig. 14 to Fig. 17 show the training, testing and checking phases respectively. Initially, the model has been developed to accurately learn the patterns of the system through previously gathered dataset with minimum error rate and high precision. Then this model was tested with unseen dataset for identification of its ability in real time environment. Results were convincing for testing phase, so finally results were gathered for checking phase also.

Fig. 18 shows the structure of ANFIS that is used to train the model used to identify the impact of input variables over output. As the structure indicates that it is a fully connected model in which we will analyze the impact of tiny variation in inputs on output.

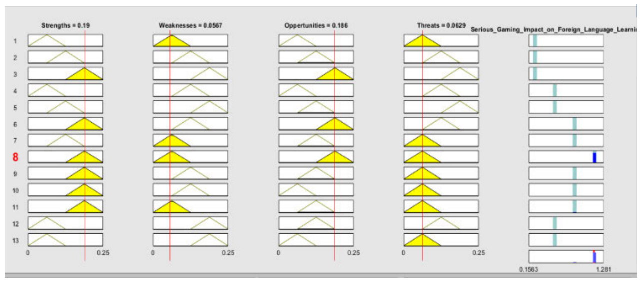

FIGURE 9. Influence of input variables on output variable.

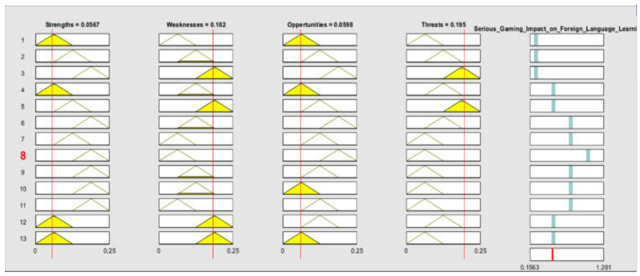

FIGURE 10. Influence of input variables on output variable.

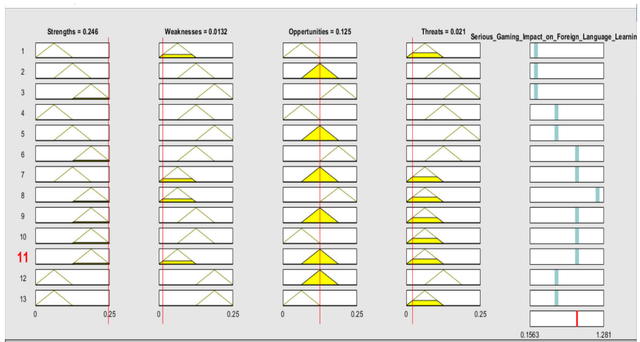

FIGURE 11. Influence of input variables on output variable.

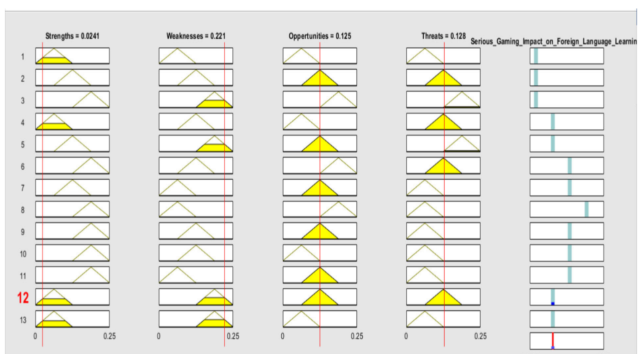

FIGURE 12. Influence of input variables on output variable.

IV. EXPERIMENTAL RESULTS

Several trials were carried out, and the sizes of the training, testing and checking data sets were assessed by carrying the classification precision into account.

The data was split into three sets: the training data set, testing dataset and the checking data set. The training data set is being used to train the ANFIS, while testing and checking data set are used to validate the performance and robustness of

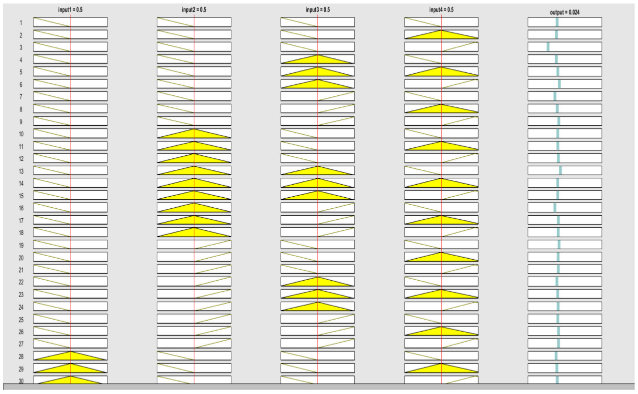

FIGURE 13. Influence of input variables on output variable.

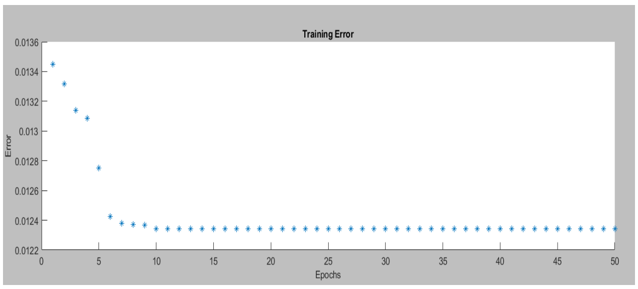

FIGURE 14. Training model specifying error rate on certain epochs.

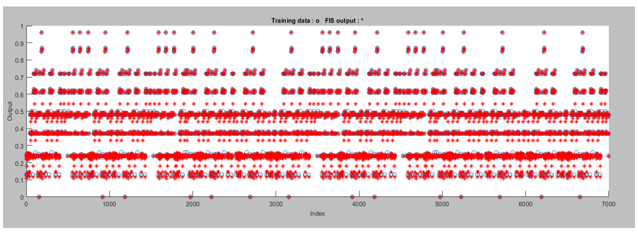

FIGURE 15. Mapping of training data (o) Against FIS output (+).

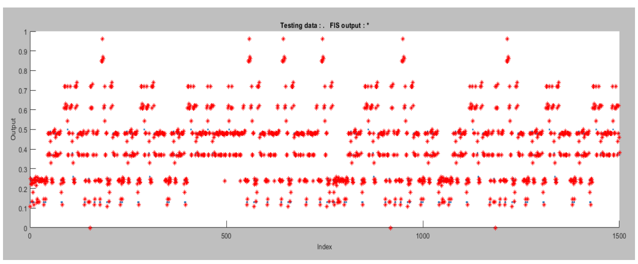

FIGURE 16. Mapping of testing data against FIS output (*).

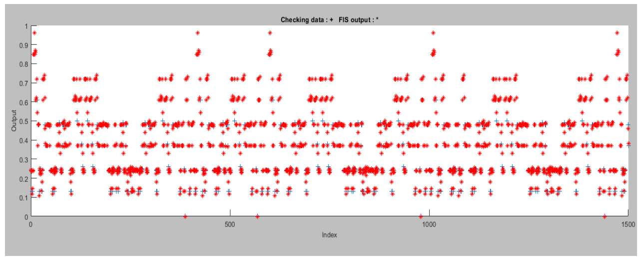

FIGURE 17. Mapping of checking data (+) against FIS output (*).

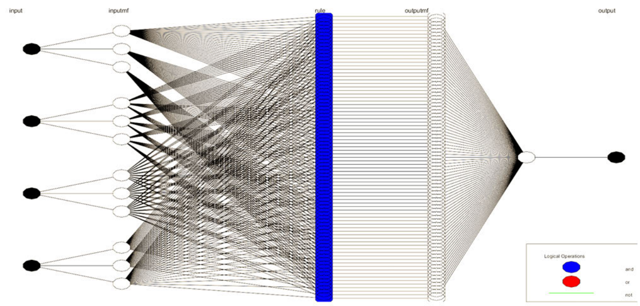

FIGURE 18. Structure of ANFIS to simulate input-output.

TABLE 8. Rule base.

If (strengths are moderate) and (weaknesses are moderate) and (opportunities are moderate) and (threats are moderate) then (Impact is neutral) (1)
If (strengths are moderate) and (weaknesses are minimal) and (opportunities are moderate) and (threats are minimal) then (Impact is positive effect) (1)
If (strengths are minimal) and (weaknesses are moderate) and (opportunities are minimal) and (threats are high) then (Impact is negative effect) (1)
………………………………
If (strengths are High) and (weaknesses are none) and (opportunities are high) and (threats are none) then (highly influential) (1)

TABLE 9. Configuration of ANFIS.

Name='Impact Analysis'	Type='Sugeno'	Version=2.0	NumInputs=4
NumOutputs=1	NumRules=81	AndMethod='prod'	OrMethod='probor'
ImpMethod='prod'	AggMethod='sum'	DefuzzMethod='wtaver'	OptiMethod='hybrid'

the trained ANFIS model for prospects adaptation. We need to analyze a variety of characteristics that carry significant roles in shaping this model in identifying the optimal model to deal with the problem of prospect identification.

The best ANFIS configuration will be determined by comparing RMSE, MSE, and MAE values for various epoch numbers also the number of membership functions delegated to each ANFIS structure. In connection to ANFIS model training, five major factors will be considered: overfitting, the number of membership functions, the form of membership functions, training data size and epoch number. We analyzed the dataset using various ANFIS configurations, and various metrics were created each time, but we concentrated on the most important findings in this analysis. The best configurations results are presented here with the assistance of Table 10.

There are four inputs as well as membership function number (3 3 3 3) in Configuration$_{A,}$. Configuration$_{B}$ and Configuration$_{C}$. The configurations were designed around

TABLE 10. Configuration of ANFIS.

ANFIS Configuration	*Configuration$_A$*			*Configuration$_B$*			*Configuration$_C$*		
Number of Inputs	4								
Membership Functions Type	Gausian Curve			Pi-Shaped			Triangular Shaped		
Number of Membership Functions	3*3*3*3								
Training Dataset	3500	5250	7000	3500	5250	7000	3500	5250	7000
Testing Dataset	1500	1125	1500	1500	1125	1500	1500	1125	1500
Checking Dataset	1500	1125	1500	1500	1125	1500	1500	1125	1500
Epoch Number	25	40	50	25	40	50	25	40	50
Number of Nodes	193			193			193		
Number of Linear Parameters	405			405			405		
Number of Non-Linear Parameters	24			48			36		
Total Number of Parameters	429			453			441		
Number of Fuzzy Rules	81			81			81		
Input Combinations	Strengths, Weeknesses, Oppertunities, Threats								
RMSE(Training)	**0.019653**	0.009644	0.005449	0.012522	0.010423	0.010156	0.0023415	0.0022381	**0.002341**
RMSE(Testing)	**0.019999**	0.009956	0.005786	0.012578	0.010984	0.010265	0.0022615	0.0022784	**0.002643**
RMSE(Checking)	**0.038678**	0.021378	0.018437	0.033532	0.030217	0.022136	0.00439616	0.00239231	**0.0129711**
MSE(Training)	**0.00039**	0.00009	0.00003	0.00046	0.00011	0.0001	0.00005	0.00003	**0.00001**
MSE(Testing)	**0.0004**	0.000099	0.000033	0.000158	0.000121	0.000105	0.000005	0.000005	**0.000007**
MSE(Checking)	**0.0015**	0.00046	0.00034	0.00112	0.00091	0.00049	0.00002	0.00001	**0.00017**
MAE(Training)	**0.015657**	0.005674	0.002443	0.009512	0.007418	0.0071126	0.0020419	0.0019389	**0.001943**
MAE(Testing)	**0.015899**	0.005965	0.002664	0.009912	0.007613	0.0073125	0.0020549	0.0019657	**0.001987**
MAE(Checking)	**0.035669**	0.018364	0.015435	0.030561	0.027245	0.019128	0.00137856	0.00209224	**0.0099734**

four inputs that were fed into a network. The inputs were chosen using a SWOT analysis to determine the effect of the obtained inputs on the output format for identifying foreign language learning prospects through serious games. Strengths, Weaknesses, Opportunities, and Threats were the four input parameters that were monitored.

We shall start by testing our Configuration$_{A,}$. Configuration$_B$ and Configuration$_C$ with default membership function numbers (3 3 3 3), and membership function types ''Gaussian, Pi-Shaped, and Triangular-Shaped'' respectively. These configurations include membership functions for each of the four inputs, resulting in a total of twelve membership functions. There are 81 fuzzy rules in the generated fuzzy inference system configuration.

Table 10 shows that the Triangular-shaped (trimf) membership function works efficiently with the smallest RMSE, MSE and MAE during validation with the epoch numbers 25, 40, and 50, while the Pi-shaped (pimf) membership function provided the worst results. Table 10 summarizes the membership function numbers and types that were evaluated, along with their corresponding epoch numbers.

The experimental findings presented in table below reveal that the chosen fuzzy rule base delivers accurate findings for the dataset used in the training, testing and checking processes, with sample sizes of (3500, 5250, 7000), (1500, 1125, 1500) and (1500, 1125, 1500) respectively. However, not all scenarios can be addressed during the fuzzy rule-base development process, and ANFIS detects some essential rules. Tables 10 demonstrate that the collection of 81 rules covers all possible scenarios for all input attributes; though, a few of the rules have been built to demonstrate irrational decisions because the training sample size was insufficient to cover all potential scenarios.

As shown in Fig. 19, the ANFIS configurations are evaluated based on their success in training and checking sets. In terms of data sample, number of membership functions, and type of membership functions, the ANFIS models have shown substantial performance variations against the evaluation criteria. The ANFIS models tend to be reliable and consistent in various subsets, where all of the RMSE, MSE, and MAE values are smaller. It also indicates that the lowest values of RMSE, MSE, and MAE for training, testing and checking are **(0.002341,0.002643, 0.0129711), (0.00001**, **0.000007**, **0.00017)** and (**0.001943**, **0.001987**, **0.0099734)** respectively. Similarly, the highest RMSE, MSE, and MAE values for training and testing are **(0.019653, 0.019999, 0.038678)**, (**0.00039**, **0.0004**, **0.0015) and (0.015657, 0.015899, 0.035669)** respectively. As a result, ANFIS based Configuration$_C$ was chosen as the best-fit model for identifying foreign language learning prospects through serious games.

The suitability and usability of the ANFIS method was explored in this paper using a variety of data sets, membership functions, and membership function types. For the purpose of identifying learning prospects, different ANFIS configurations were generated using a combination of inputs, sample training data, membership function numbers, and membership function types. These ANFIS models were developed and put to the test. The ANFIS models and observation findings were correlated and evaluated. The performance of ANFIS models in training and checking data sets was compared.

One of the motivations for developing an adaptive learning prospects identification scheme was to assist the administration in tracking progress during this whole activity, which entails: i) recording the participants' context during SWOT

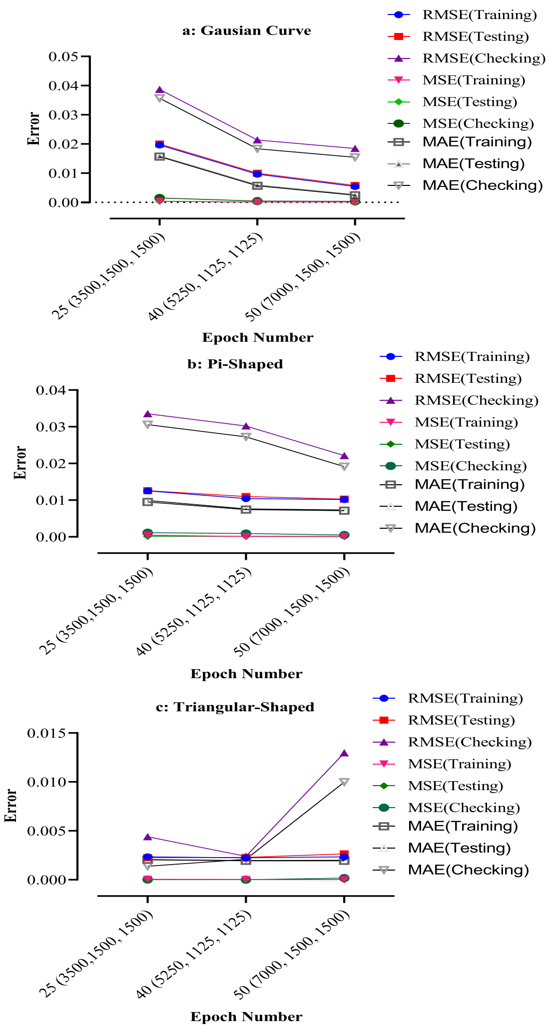

FIGURE 19. Configuration$_A$, Configuration$_B$ and Configuration$_C$ error rate.

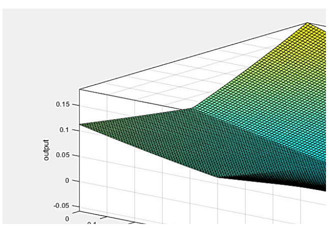

FIGURE 20. Impact of strengths and weaknesses on learning foreign language through serious games.

analysis and then ii) using the recorded participants' context to develop a learning prospects identification model. If the recorded participants' context, which shapes the learning prospects identification model, accurately reflects the entire operation and its changes constantly using ANFIS as a reasoning engine, then the identified model can maintain pace with the shift in participants' context to identify prospects by making precise decisions.

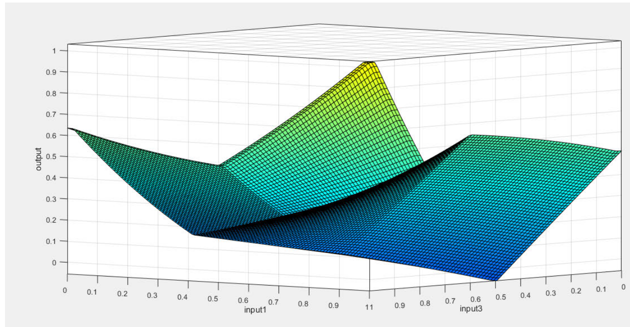

FIGURE 21. Impact of strengths and opportunities on learning foreign language through serious games.

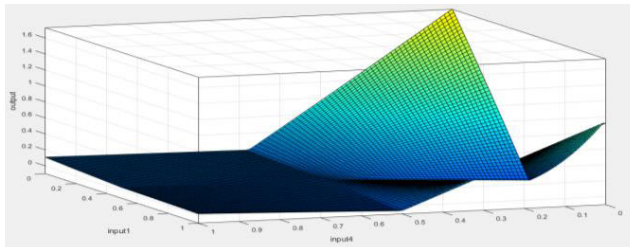

FIGURE 22. Impact of strengths and threats on learning foreign language through serious games.

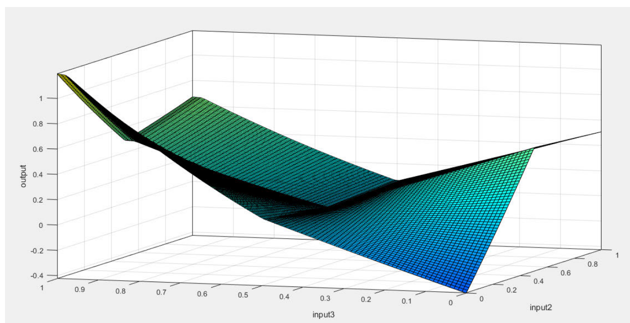

FIGURE 23. Impact of weaknesses and opportunities on learning foreign language through serious games.

Ultimately, to better understand and analyze the impact of any two input variables on the output variable, we can use surface viewer given in Fig. 20 to Fig. 25, that provides insights about different input-output patterns in different situations.

With a change in the 'Strengths' value, the 'prospects of foreign language learning through serious games' change fairly shown in Fig. 20. Changes in the input 'Weaknesses' have an inverse relationship with the output 'Prospects of Foreign Language Learning through Serious Games.' This implies that the high value of input 'Weaknesses' reduces the chances of learning.

Similarly, in Fig. 21, a change in the combined effect of input values 'Strengths' and 'Opportunities' boosts the output 'prospects of foreign language learning through serious games' by leveraging the individual positive impact of both

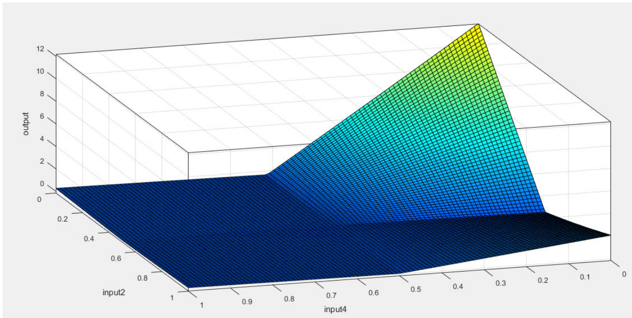

FIGURE 24. Impact of weaknesses and threats on learning foreign language through serious games.

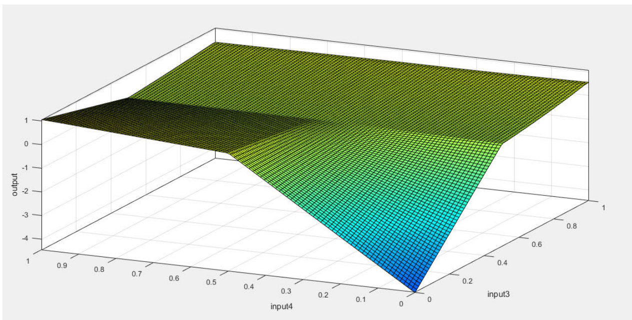

FIGURE 25. Impact of opportunities and threats on learning foreign language through serious games.

parameters and also positive relationship between the two parameters on learning prospects.

Likewise, in Fig. 24, the pair of inputs is 'Weaknesses' and 'Threats' their high values reduces the value of output 'prospects of foreign language learning through serious games' by taking into account the negative impact of both parameters on learning prospects.

Analogously, all other pairings of inputs from 'Strengths', 'Weaknesses', 'Opportunities', and 'Threats' represent a direct or inverse relationship with the output 'Prospects of Foreign Language Learning through Serious Games' as depicted in the all surface views below.

The effectiveness of ANFIS is evaluated using surface images of various inputs, such as 'Strengths,' 'Weaknesses,' 'Opportunities,' and 'Threats,' versus 'Prospects of Foreign Language Learning through Serious Games'. The amount of error describes the estimation's efficacy and in our case the amount of error for the estimation is too small.

V. DISCUSSION AND CONCLUSION

Serious games are invading our daily life and are taking a place in educational systems. The mitigating opinions about their efficiency as a learning material make them as an object of diverse points of views from criticism to support. First, serious games in language learning will bring joy and fun in classrooms, which could diminish stress and boredom. Serious games could be seen as an evolution in the educational system and fitting new generations' requirements and needs. Second, they constitute a non-evitable change that researchers must study and analyze in order to prevent its impact and effects on learning outcomes and levels. Regarding the heavy content of a foreign language learning, the use of serious game has many advantages and opportunities. The customizable environment of a game makes students more comfortable when working in a pleasant and flexible environment that they can control. Students seem to appreciate the use of such new material in classrooms, the adoption of serious games is expected to increase in the future. Ludic games are able to attract student's attention and make them feel engaged and involved in the learning process. A new challenging and motivating experience which could resolve the main issues of lack of concentration and distraction in classrooms. They help shy students to perform better since there will be no social or gender discrimination and students will have the right to error without getting mocked by their classmates.

However, many threats and drawbacks still hyping users and researchers. In fact, games could bring lack of control and discipline issues in classrooms. Teachers could lose the control about what every student is doing in class. Students may have a negative perception about learning by the means of games. They could perceive it as childish or immature. Learners may not apprehend the wisdom behind such learning process and feel frustrated to use it if the game is inadequate to their learning style or level. Students require that proposed game should consider several parameters regarding their gender, their foreign language level, their major gaps and their mother tongue. They value a strong connection between the efficiency of learning and the extent to which the game is able to be flexible with their own profile. These insights are a first step towards an optimal use of serious games to afford a valuable foreign learning experience and it helps for better designing of serious games.

Data analysis allowed us to identify a set of constructs that seem to be relevant for effective use of serious games in the context of foreign language learning. Identified parameters may positively or negatively influence the effectiveness of serious games. The factors, such as engagement, motivation, learning style and customization have been already proven by academic literature to be relevant in the context of educational serious games [34], [36], [41]. These factors are essential for effective use of serious gaming, and all have an obvious impact by enabling or inhibiting students when learning in a ludic environment.

The statistics obtained through data analysis are relatively discrete and cannot present the overall impact of the involved variables of SWOT analysis. To serve this purpose there is a dire need of expert system that can precisely predict the overall influence of integrated variables. It would be helpful to identify the viability of serious games for foreign language learning purpose. So, the proposed prospects identification system will analyze the impact of serious games on learning foreign language by considering smallest to largest variations in strengths, weaknesses, opportunities and threats. The approach based on Fuzzy Logic and Neural Networks found to be very effective and operative under different rules and conditions.

To implement a microscopic technique that would consider tiniest change in the inputs for the prospects identification model and later for its calibration. Our idea constitutes on an online calibration approach that means if the prospects of learning are not productive then what is the best possible configuration of inputs that can improve the prospects in real time. Our suggestion is followed by a general presentation of the simulation and calibration of microscopic viewpoints, with special reference to the unique scenario of serious gaming effectiveness in foreign language learning.

The proposed approach has been evaluated by simulation findings and error during implementation in conjunction with the framework responsible for the calibration through the ANFIS (MATLAB R2020a). This study's input data consisted of feedback from current students of Information Systems via SWOT analysis. Once again, Takagi-Sugeno-Kang FIS has shown its effectiveness in adaptive systems by improving the parameter offsetting mechanism. It is a hybrid model that appropriately utilizes the upsides of Mamdani and Takagi–Sugeno–Kang (TSK) fuzzy structures in a distinctive learning process, leading towards more robustness and better accuracy of the extracted TSK models. Based on a Mamdani fuzzy structure, the primary adaptive granulation process provides better generalization, more compactness, and a reasonable initial solution for the secondary optimization stage. The algorithm's fine-tuning strategy, on the other hand, is built on a TSK structure, which provides a lower process load, universal approximation property, and the possibility of using more appropriate optimization strategies, resulting in better model accuracy.

The error rate during training, testing and checking procedures suggested that the hybrid solution may put a model closer to the actual model. A hybrid solution has the great advantage of having the time-varying reward centered on varying inputs found in SWOT analysis. In addition, this idea implies the specific interaction between learning foreign language and serious games according to the microscopic prospects modeling technique.

Further studies should expand this method at the mesoscopic modeling stage of prospects, where the criteria for assessment of foreign language learning prospects can be assigned to groups of students instead of individual students. The contact restriction guarantee within a group of students for transparency would be a major challenge in that direction because mesoscopic modeling does not provide adequate granularity relative to models of microscopic prospects identification.

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